

Lecture 9: State Feedback & Observer

©Angela Fontan, angfon@kth.se

Dept. Decision and Control Systems, EECS, KTH Royal Institute of Technology, Sweden

Based on slides by ©Bo Wahlberg

Disclaimer: These (informal) notes are not intended to replace the textbook or the material contained therein that students are expected to cover and study. If you see any typos, please let me know.

1 Outline

In the previous lectures (F7, F8)..

- State-space representation:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t), & x(0) = x_0 \\ y(t) = Cx(t) + Du(t) \end{cases} \quad (1)$$

- $x(t) \in \mathbb{R}^{n \times 1}$ state vector ($n \geq 2$)
- $u(t) \in \mathbb{R}^{m \times 1}$ input vector
- $y(t) \in \mathbb{R}^{p \times 1}$ output vector
- $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, $D \in \mathbb{R}^{p \times m}$ (usually: $D = 0$)

- Observability and Controllability (observable and controllable states)

Today: We use these foundations for *control design*: given the state (or an estimate of it), we specify a feedback law that places the closed-loop poles at desired locations. We consider two scenarios:

- State feedback (from $x(t)$): the full state vector $x(t)$ is assumed to be directly measurable
- State feedback with observer (from $y(t)$): only the output $y(t)$ is measurable; the state is estimated by an observer and the estimate is used in the feedback law.

Table 1: Today's Ordlista

English	Svenska
tillstånd	state
state-space (form/representation)	tillstånds(form/beskrivning)
Pole placement	Polplacering
State feedback	Tillståndsåterkoppling
steady state/equilibrium	stationärt tillstånd
observer	observatör
estimation error	skattningsfelet
measurement error	mätfelet
complementary sensitivity function	komplementära känslighetsfunktionen
sensitivity function	känslighetsfunktionen

2 State Feedback and Pole Placement

2.1 The State Feedback Law

Assume that the full state vector $x(t)$ is available for measurement. A *state feedback* law is defined as

$$u(t) = -Lx(t) + L_0 r(t), \quad (2)$$

where $L \in \mathbb{R}^{m \times n}$ is the *feedback gain matrix* and $L_0 \in \mathbb{R}^{m \times p}$ is a (prefilter/reference) gain chosen to achieve zero steady-state error. Throughout most of this course, L_0 is a scalar $\ell_0 \in \mathbb{R}$, as we primarily consider systems with $m = p = 1$.

Assumption 2.1. In F9, we assume:

- $m = p = 1$, implying: $L_0 = \ell_0 \in \mathbb{R}$, $B \in \mathbb{R}^{n \times 1}$, $C \in \mathbb{R}^{1 \times n}$, and $L \in \mathbb{R}^{1 \times n}$;
- $D = 0$.

Substituting (2) into (1) (with $D = 0$) yields the *closed-loop system*:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \\ u = -Lx + \ell_0 r \end{cases} \Rightarrow \begin{cases} \dot{x} = Ax + B(-Lx + \ell_0 r) \\ y = Cx \end{cases} \Rightarrow \boxed{\begin{cases} \dot{x} = \underbrace{(A - BL)}_{A_c} x + \underbrace{B \ell_0}_{B_c} r, \\ y = \underbrace{C}_{C_c} x. \end{cases}} \quad (3)$$

The closed-loop matrices are $(A_c, B_c, C_c, D_c) = (A - BL, B\ell_0, C, 0)$ ¹.

The *poles of the closed-loop system* are given by the eigenvalues of $A_c = A - BL$, i.e., the roots of the characteristic equation

$$\det(sI - (A - BL)) = 0.$$

2.2 Pole Placement

We observe that:

- The matrix $A - BL$ has n eigenvalues (= poles of closed-loop system)
- The gain matrix L has n free parameters (for a single-input system, $m = 1$)

The following fundamental result holds.

Theorem 2.2 (Pole Placement, Resultat 9.1). *If the system (A, B) is controllable, then the poles of the closed-loop system can be placed arbitrarily, i.e., the eigenvalues of $A - BL$ can be assigned to any desired set of n locations in the complex plane by appropriate choice of L .*

The designer's freedom to place the poles anywhere is, however, limited in practice by the magnitude of the resulting control signal $u(t)$: placing poles far to the left in the complex plane demands large gains in L , which may saturate actuators².

Remark 2.3. If the system is in controllable canonical form (check F7 and Section 2.5), then calculating L is very easy!

¹**Homework:** Draw the block diagram!

²**Homework:** Study Exempel 9.6 on LQ-control!

Example 0: Pole placement for the double integrator

Consider the following system $\Sigma = (A, B, C, D)$ and feedback law u (i.e., $L = [l_1 \ l_2]$ and ℓ_0):

$$\begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} x \\ u = -\begin{bmatrix} l_1 & l_2 \end{bmatrix} x + \ell_0 r \end{cases}$$

The closed-loop state matrix is

$$A - BL = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} l_1 & l_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -l_1 & -l_2 \end{bmatrix},$$

with characteristic polynomial

$$\det(\lambda I - (A - BL)) = \det \begin{bmatrix} \lambda & -1 \\ l_1 & \lambda + l_2 \end{bmatrix} \stackrel{a}{=} \begin{vmatrix} \lambda & -1 \\ l_1 & \lambda + l_2 \end{vmatrix} = \lambda^2 + l_2 \lambda + l_1.$$

Now assume, for example, that we want to place the poles at -3 and -3 (double pole). To place both poles at -3 , we match the coefficients of the characteristic polynomial with

$$(\lambda + 3)^2 = \lambda^2 + 6\lambda + 9,$$

which yields

$$l_1 = 9, \quad l_2 = 6 \quad \implies \quad u = -9x_1 - 6x_2 + \ell_0 r.$$

This gives us the closed-loop system

$$\begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 \\ -9 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \ell_0 r \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases}$$

Placing the poles instead at -3000 (approx. 1000 times faster) would require $l_1 = 9 \times 10^6$ and $l_2 = 6 \times 10^3$, illustrating the actuator-saturation constraint (= limited by the size of $u(t)$!) imposed by aggressive pole placement.

^aSometimes the determinant is written with $|\cdot|$

2.3 Choice of the Gain ℓ_0

The gain ℓ_0 is chosen so that the closed-loop transfer function satisfies $G_c(0) = 1$, i.e., the steady-state output equals the reference when $r(t) = \bar{r}$ is constant. At steady state, $\dot{x} = 0$, so from (3):

$$0 = (A - BL)\bar{x} + B\ell_0\bar{r}, \quad \bar{y} = C\bar{x}.$$

Solving for $\bar{x}$ and requiring $\bar{y} = \bar{r}$ determines ℓ_0 .

Example 1: Gain for the double integrator, cont'd from Example 1

With poles at -3 , the closed-loop system is

$$\begin{cases} \dot{x} &= \begin{bmatrix} 0 & 1 \\ -9 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \ell_0 r \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases}$$

At equilibrium with constant reference $r(t) = \bar{r}$, we have:

$$\begin{cases} r(t) &= \bar{r} \\ \dot{x} &= 0 \end{cases} \implies \begin{cases} 0 = \dot{x}_1 = x_2 \\ 0 = \dot{x}_2 = -9x_1 - 6x_2 + \ell_0 \bar{r} \end{cases} \implies \begin{cases} x_2 = 0 \\ x_1 = \frac{\ell_0}{9} \bar{r} \end{cases}$$

Since $y = x_1 = \frac{\ell_0}{9} \bar{r}$, the condition $y = \bar{r}$ requires $\ell_0 = 9$. The complete state feedback law is therefore

$$u = -9x_1 - 6x_2 + 9r.$$

Remark 2.4. The weakness of this method for determining ℓ_0 is that it requires knowledge of $G(0)$ and assumes that no disturbances affect the system. $\implies$ Introduce integral control!

2.4 Summary of State Feedback

For the system (1)

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases}$$

with transfer function

$$G(s) = \frac{b_1 s^{n-1} + \dots + b_n}{s^n + a_1 s^{n-1} + \dots + a_n},$$

the state feedback law $u = -Lx + \ell_0 r$ yields the following closed-loop system and transfer function

$$\begin{cases} \dot{x} &= (A - BL)x + B\ell_0 r \\ y &= Cx \end{cases} \implies G_c(s) = \frac{(b_1 s^{n-1} + \dots + b_n) \ell_0}{s^n + \alpha_1 s^{n-1} + \dots + \alpha_n},$$

where the coefficients α_i are determined by the chosen pole locations:

$$\det(sI - (A - BL)) = s^n + \alpha_1 s^{n-1} + \dots + \alpha_n = 0$$

Two important observations follow:

- The feedback gain L affects only the *poles* of the closed-loop system, not its zeros.
- The gain ℓ_0 affects the static gain but leaves both poles and zeros unchanged (indeed, $Y(s) = \tilde{G}_c(s) \ell_0 R(s)$, where $\tilde{G}_c(s) = \frac{b_1 s^{n-1} + \dots + b_n}{s^n + \alpha_1 s^{n-1} + \dots + \alpha_n}$).

2.5 Controllable Canonical Form

When the system is expressed in the controllable canonical form (Resultat 8.2, Exempel 9.2), the computation of L is particularly easy. In that representation,

$$A = \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_n \\ 1 & 0 & \cdots & 0 \\ \vdots & \ddots & & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad C = [b_1 \quad \cdots \quad b_n].$$

The closed-loop matrix $A - BL$ has the same companion structure (= controllable canonical form), with the 1st row replaced by $[-(a_1 + l_1) \cdots -(a_n + l_n)]$:

$$A - BL = \begin{bmatrix} -\overbrace{(a_1 + l_1)}^{\alpha_1} & -\overbrace{(a_2 + l_2)}^{\alpha_2} & \cdots & -\overbrace{(a_n + l_n)}^{\alpha_n} \\ 1 & 0 & \cdots & 0 \\ \vdots & \ddots & & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}$$

Then:

- The *desired* characteristic polynomial $s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n = 0$ is matched simply by setting $l_i = \alpha_i - a_i$ for $i = 1, \dots, n$. The zeros (encoded in C) are unchanged, confirming that state feedback cannot alter the open-loop zeros.
- The choice of ℓ_0 follows directly from $G_c(0) = 1$, i.e., $\ell_0 = \alpha_n / b_n$.

Weakness. This static choice of ℓ_0 requires exact knowledge of α_n and b_n , and provides no rejection of constant disturbances or model errors. The solution is to augment the system with integral action.

3 Integral Action via State Augmentation

To achieve zero steady-state error in the presence of disturbances and model uncertainties, an *integrator* is introduced on the tracking error $e(t) = r(t) - y(t)$. An additional state is defined as

$$x_{n+1}(t) = \int_0^t e(\tau) d\tau = \int_0^t (r(\tau) - y(\tau)) d\tau, \quad (4)$$

so that $\dot{x}_{n+1} = r - y = r - Cx$. The original state-space model is then *augmented* as

$$\underbrace{\begin{bmatrix} \dot{x} \\ \dot{x}_{n+1} \end{bmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x \\ x_{n+1} \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{\mathbf{B}} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r,$$

and the augmented state feedback law is

$$u = -Lx - l_{n+1}x_{n+1} =: -\mathbf{L}\mathbf{x}.$$

An illustration is given in Fig. 1. The gain vector $\mathbf{L}$ is chosen so that $\mathbf{A} - \mathbf{B}\mathbf{L}$ has $n + 1$ eigenvalues at the desired closed-loop pole locations. At steady state,

$$\begin{cases} \dot{x} &= 0 \\ \dot{x}_{n+1} &= r - y = 0 \end{cases}$$

hence, $\dot{x}_{n+1} = 0$ implies $y = r$ exactly, regardless of constant step disturbances or model errors (as expected from the integral action!).

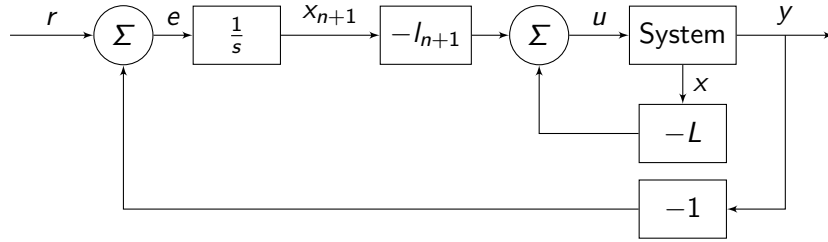


Figure 1: Block diagram: Integral action

4 Observer Design

4.1 Motivation

State feedback requires that all components of $x(t)$ be measured. In practice, this is rarely the case: only the output $y(t) = Cx(t)$ is directly accessible. An *observer* (or state estimator) reconstructs $x(t)$ from the known quantities $u(t)$ and $y(t)$.

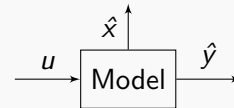
Example 2: Open-Loop Simulation and the Need for Feedback

Let's assume we have a model:

$$\begin{cases} \dot{x} = Ax + Bu & x(0) = x_0 \\ y = Cx \end{cases}$$

A naive approach is to simulate the system model in parallel:

$$\begin{cases} \dot{\hat{x}} = A\hat{x} + Bu & \hat{x}(0) = \hat{x}_0 \\ \hat{y} = C\hat{x} \end{cases}$$



The difference $y(t) - \hat{y}(t)$ measures the simulation error. However, if $\hat{x}(0) \neq x_0$, the estimation error $\tilde{x}(t) = x(t) - \hat{x}(t)$ satisfies $\dot{\tilde{x}} = A\tilde{x}$, which converges to zero only if A is stable, a property that cannot be relied upon in general. The solution is to close a feedback loop around the error signal $y - \hat{y}$.

4.2 The (Luenberger) Observer

Adding proportional feedback of the output error $y - \hat{y}$ to the simulated dynamics yields the *Luenberger observer*:

$$\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x}) = (A - KC)\hat{x} + Bu + Ky, \quad (5)$$

where $K \in \mathbb{R}^{n \times p}$ is the *observer gain matrix* (for a single-output system, K is a column vector). The observer is a dynamical system driven by both the known input $u(t)$ and the measured output $y(t)$.



4.3 Estimation Error Dynamics and Observer Pole Placement

The *estimation error* $\tilde{x}(t) = x(t) - \hat{x}(t)$ evolves according to:

$$\dot{\tilde{x}} = \dot{x} - \dot{\hat{x}} = Ax + Bu - (A\hat{x} + Bu + K(Cx - C\hat{x})) = (A - KC)\tilde{x}.$$

Its solution is

$$\tilde{x}(t) = e^{(A-KC)t}\tilde{x}(0),$$

where $\tilde{x}(0)$ is the initial error. Therefore:

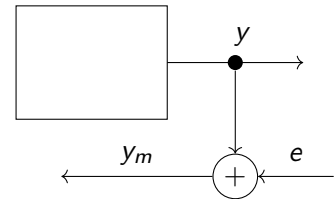
- $\tilde{x}(t) \rightarrow 0$ as $t \rightarrow \infty$ if and only if all eigenvalues of $A - KC$ lie in the open LHP.
- The rate of convergence (how quickly $\tilde{x} \rightarrow 0$) is determined by the eigenvalues of $A - KC$, which can be freely placed by choice of K , provided the system is *observable*.

Theorem 4.1 (Observer Pole Placement, Resultat 9.2). *If the system (C, A) is observable, then the eigenvalues of $A - KC$ can be assigned to any desired locations by appropriate choice of K .*

4.4 Effect of Measurement Noise

In practice, the measured output is corrupted by noise: $y_m(t) = y(t) + e(t)$. Substituting y_m for y in (5), the estimation error satisfies

$$\dot{\tilde{x}} = (A - KC)\tilde{x} + Ke.$$



This reveals a fundamental trade-off in the choice of K :

- **Large K** : fast convergence of $\tilde{x}$ to zero, but high sensitivity to measurement noise $e(t)$.
- **Small K** : low sensitivity to noise, but slow estimation.

The optimal trade-off, minimizing a weighted combination of estimation error and noise sensitivity, is achieved by the *Kalman filter*, which computes K as the solution to a Riccati equation given statistical characterizations of the process and measurement noise.

Practical rule of thumb. The observer poles should be placed *faster* than the controller poles (“more to the left”):

$$|\text{eig}(A - BL)| \leq |\text{eig}(A - KC)|.$$

This ensures that the state estimate converges quickly enough not to degrade closed-loop performance.

Example 3: Observer design for the double integrator

For the system of Example 1 (controller poles at -3), we design the observer with poles at -4 . We compute:

$$A - KC = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} -k_1 & 1 \\ -k_2 & 0 \end{bmatrix},$$

$$\det(sI - (A - KC)) = s^2 + k_1s + k_2.$$

Matching with $(s + 4)^2 = s^2 + 8s + 16$ gives $k_1 = 8$, $k_2 = 16$. The observer equations are therefore:

$$\dot{\hat{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \hat{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 8 \\ 16 \end{bmatrix} (y - \begin{bmatrix} 1 & 0 \end{bmatrix} \hat{x}).$$

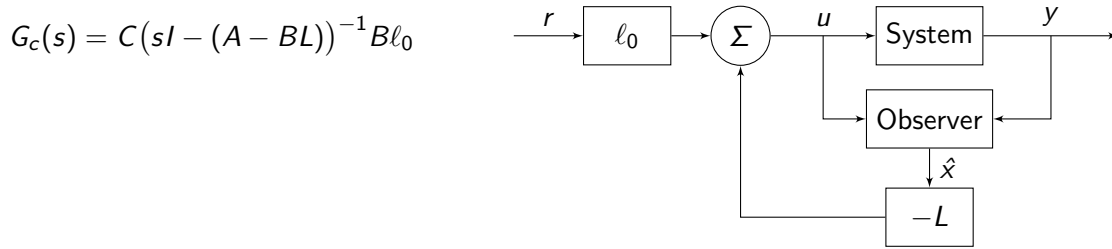
5 Output Feedback from Reconstructed States

5.1 The Separation Principle

Consider the system (1). When the full state is unavailable, the state feedback law is applied to the *estimated* state $\hat{x}$ produced by the observer:

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Bu + K(y - C\hat{x}) \\ u(t) &= -L\hat{x}(t) + \ell_0 r(t).\end{aligned}$$

The complete closed-loop system has order $2n$ (the n states of the plant and the n states of the observer).



The *separation principle* holds: the closed-loop transfer function from r to y is identical to that obtained with perfect state feedback ($G_c!$), provided $\hat{x}(0) = x(0)$. This follows from the fact that the estimation error $\tilde{x}$ satisfies an autonomous equation $(A - KC)\tilde{x}$ and is not controllable (it cannot be influenced by r), but is observable. The corresponding poles are canceled from the input-output transfer function.

In consequence, the $2n$ poles of the combined system decompose into:

- *n controller poles*: eigenvalues of $A - BL$, chosen by state feedback design. The controllable subspace has n dimension with vectors in the type $[x_0]^T$. For this reason, G_c has order n .
- *n observer poles*: eigenvalues of $A - KC$, chosen independently by observer design.

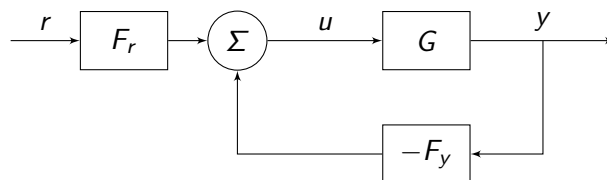
The two designs are therefore decoupled and can be performed independently.

5.2 Alternative Transfer-Function Implementation

The combined observer-plus-state-feedback system admits an equivalent implementation as a classical two-degree-of-freedom controller with a reference filter F_r and an output feedback filter F_y :

$$\begin{aligned}F_r(s) &= \left(1 - L(sI - (A - BL - KC))^{-1}B\right)\ell_0, \\ F_y(s) &= L(sI - (A - BL - KC))^{-1}K.\end{aligned}$$

This form makes explicit that the combined design is equivalent to a classical pole placement in the Laplace domain.



5.3 Robustness: Sensitivity and Complementary Sensitivity Function

The *complementary sensitivity function* of the observer-based closed-loop system is

$$T(s) = G_c(s) \cdot \underbrace{L(sI - (A - KC))^{-1}K}_{\text{observer dynamics}},$$

where $G_c(s) = C(sI - (A - BL))^{-1}B\ell_0$ is the state-feedback closed-loop transfer function. The observer gain K therefore affects not only the estimation convergence rate but also the robustness margins of the overall system. The *sensitivity function* is

$$S = 1 - T.$$

This highlights the importance of choosing K with robustness in mind (not just estimation speed) which is precisely what the Kalman filter achieves in an optimal sense.